

C# Events, Delegates, and Lambdas

The Role of Events, Delegates, and Event Handlers



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C# Events, Delegates, and Lambdas

Version Check



Version Check



This course was created by using:
- C# 12



Version Check



This course is 100% applicable to:

- C# 3 or higher
- Some demos require C# 12 or higher



Course Overview



The Role of Events, Delegates, and Event Handlers

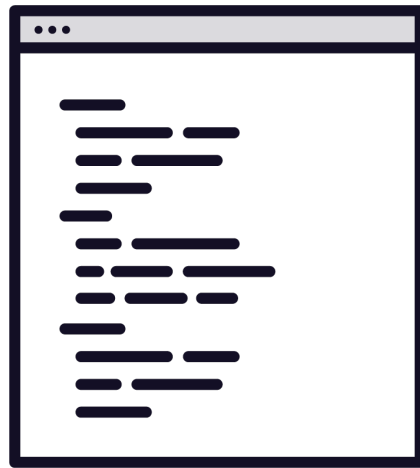
Creating Delegates, Events, and EventArgs

Handling Events

Lambdas, Action<T>, and Func<T, TResult>

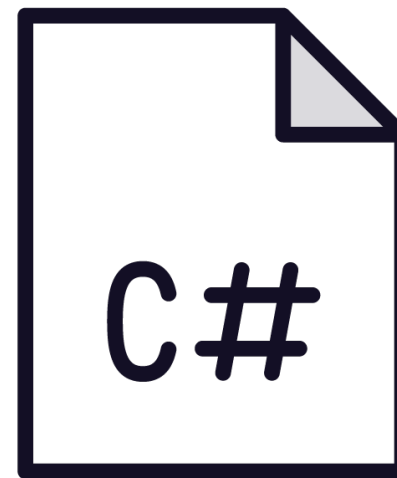


Prerequisites



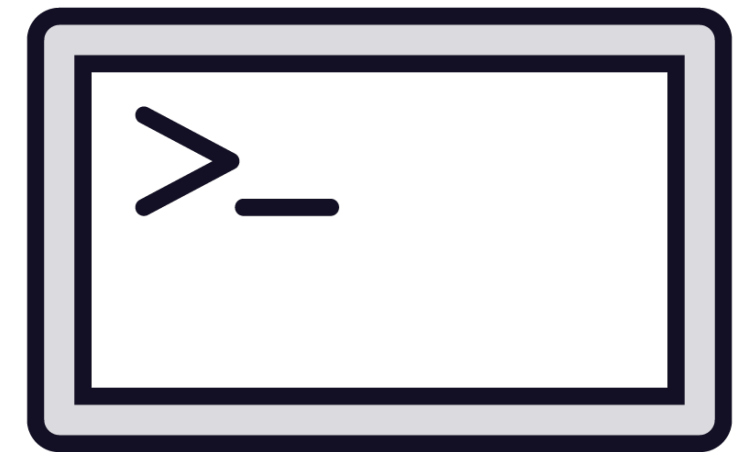
Programming

**Basic programming
and editor experience**



C#

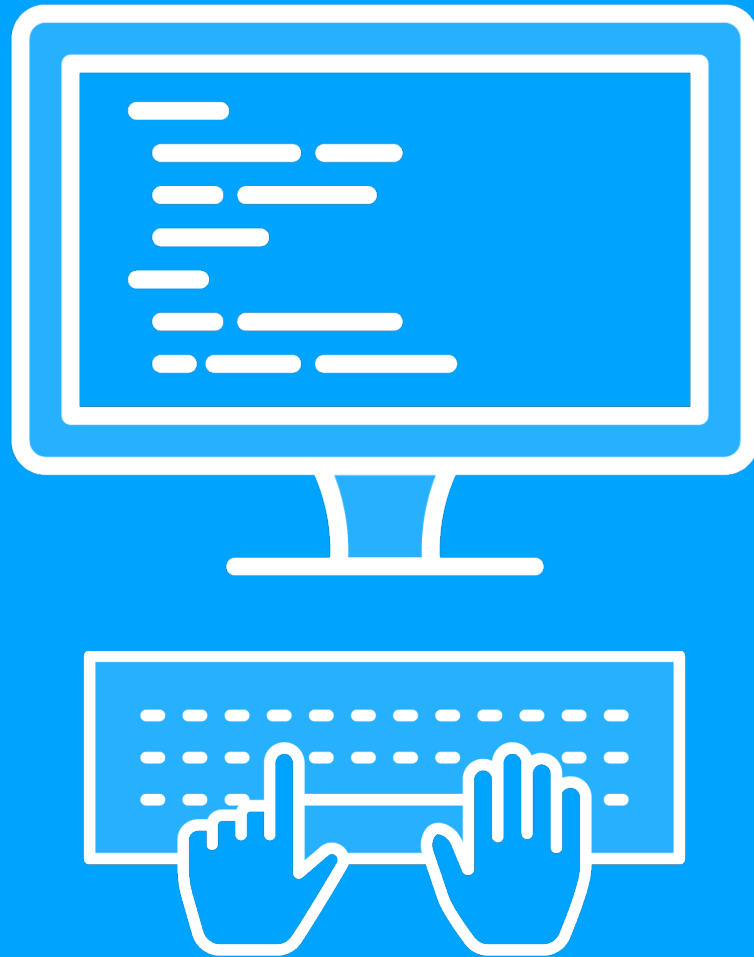
C# core concepts



Command Line

Hands-on experience





GitHub Repo

[https://github.com/DanWahlin/
Csharp-Events-Delegates-Lambdas](https://github.com/DanWahlin/Csharp-Events-Delegates-Lambdas)





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- ▶ Building and Running Your First Docker App
- ▶ Docker for Web Developers
- ▶ Kubernetes for Developers: Core Concepts
- ▶ Angular Architecture and Best Practices
- ▶ TypeScript Fundamentals
- ▶ Creating Object-oriented TypeScript
- ▶ C# Events, Delegates, and Lambdas



Overview



The Role of Events

The Role of Delegates

The Role of Event Handlers

**Events, Delegates, and Event Handlers
in Action**





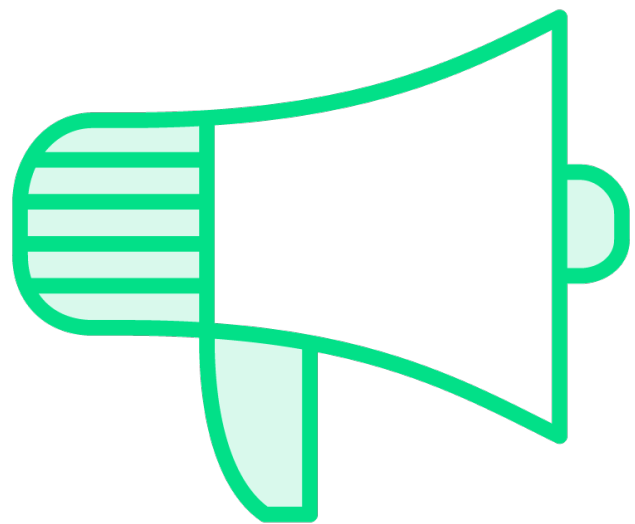
The Role of Events



Event Raiser



What Is an Event?



Events are notifications

Play a central role in C# apps

Provide a way to trigger notifications from end users or objects



The Role of Events



Events signal the occurrence of an action/notification

Objects that raise events don't need to explicitly know the object that will handle the event

Events pass EventArgs (event data)





The Role of Delegates





Delegate

What Is a Delegate?



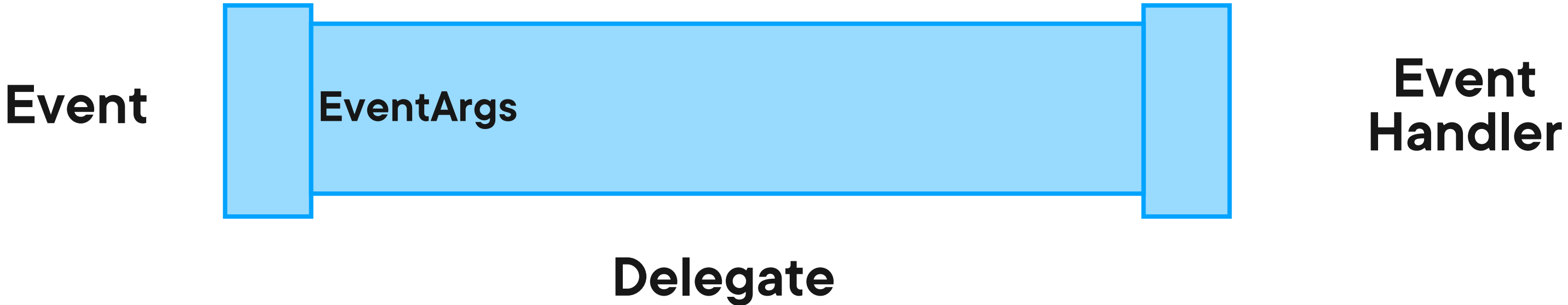
A delegate is the glue/pipeline between an event and an event handler

Specialized class based on the MulticastDelegate base class

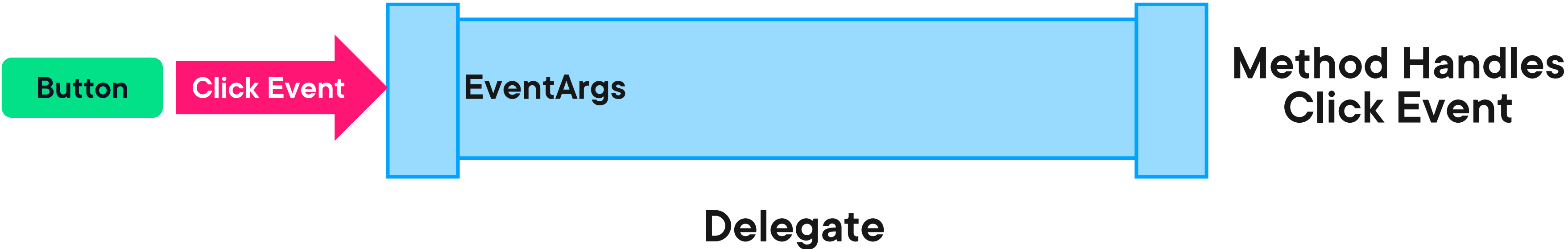
Often called a "Function Pointer"



Delegates Are a Pipeline



Delegates Are a Pipeline





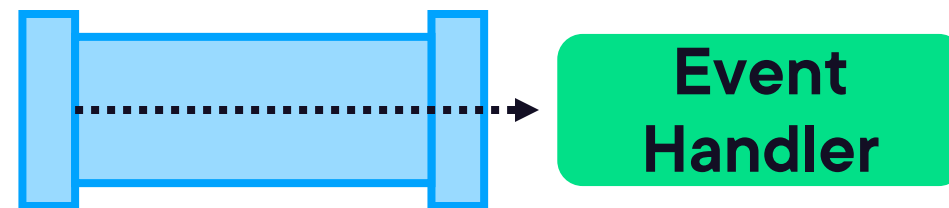
The Role of Event Handlers



Event Handler



What Is an Event Handler?



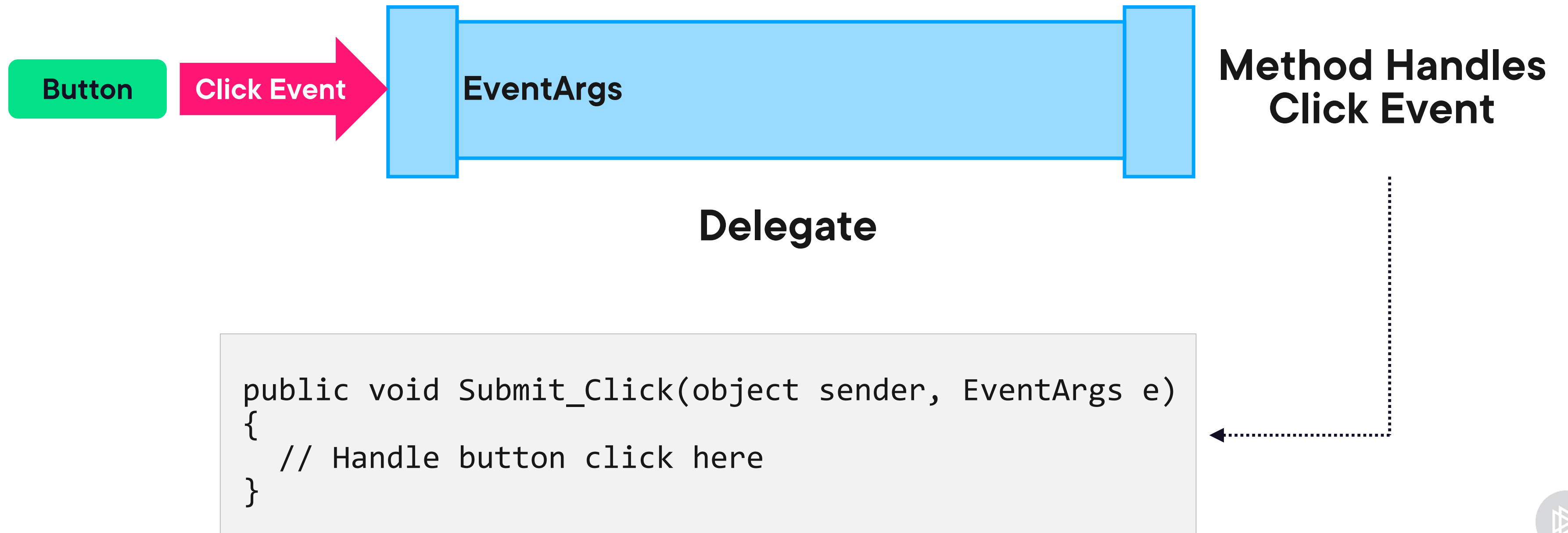
An event handler is responsible for receiving and processing data from a delegate

Event handler may receive parameters, for example:

- Sender
- EventArgs

EventArgs are responsible for encapsulating event data

Delegates Are a Pipeline





Events, Delegates, and Event Handlers in Action



Summary



Certain types of C# apps rely heavily on events and delegates

Events provide notifications and send data using EventArgs

Delegates act as the glue/pipeline between events and event handlers

Event handlers receive and process EventArgs data



Up Next:

Creating Delegates, Events, and EventArgs

